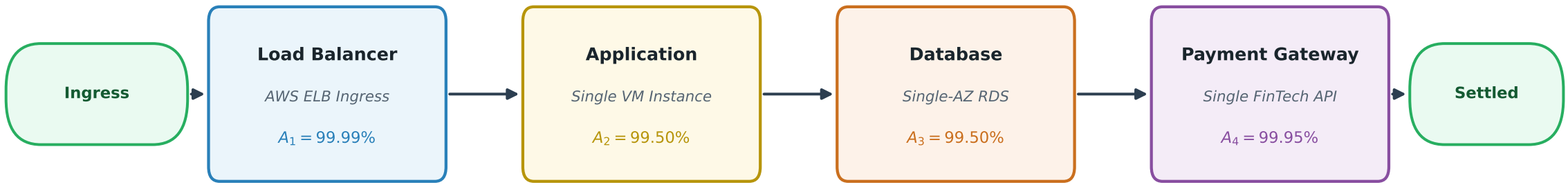
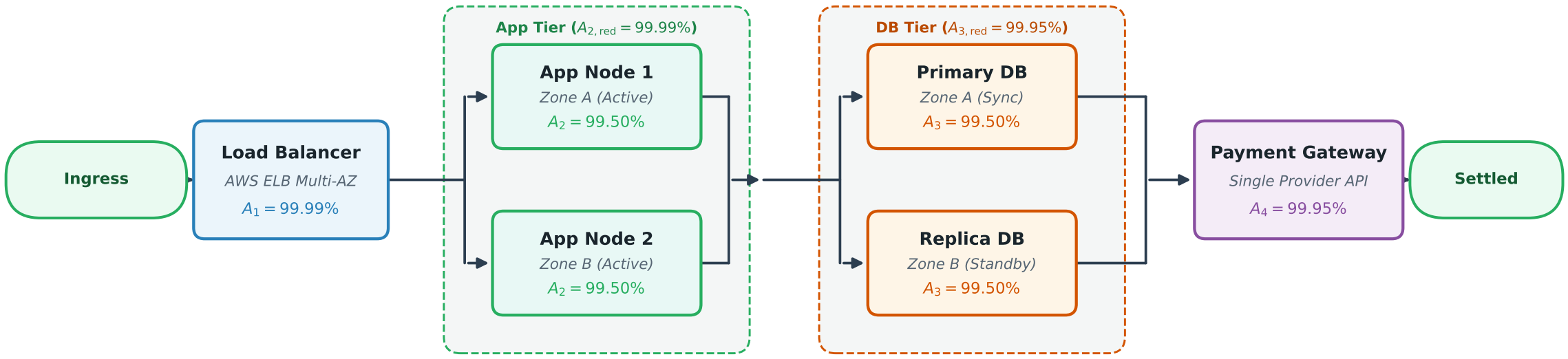
## (a) Scenario 1: Baseline Architecture (Pure Series)

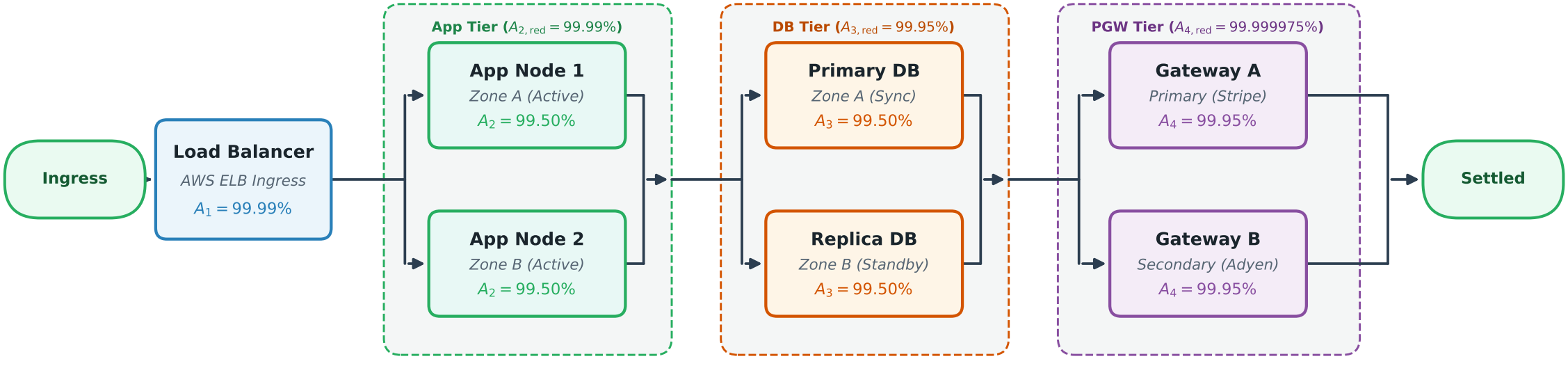
$A_{\text{base}} = 98.943\%$ (92.58 h/yr)




## (b) Scenario 2: Multi-AZ Series-Parallel Architecture

$A_{\text{red}} = 99.880\%$ (10.51 h/yr, $-88.6\%$)




## (c) Scenario 3: Multi-Gateway Expansion (Full Redundancy)

$A_{\text{scen3}} = 99.930\%$ (6.13 h/yr, $-93.4\%$)